Lecture 16: Summarizing Data

Heather Mattie, PhD

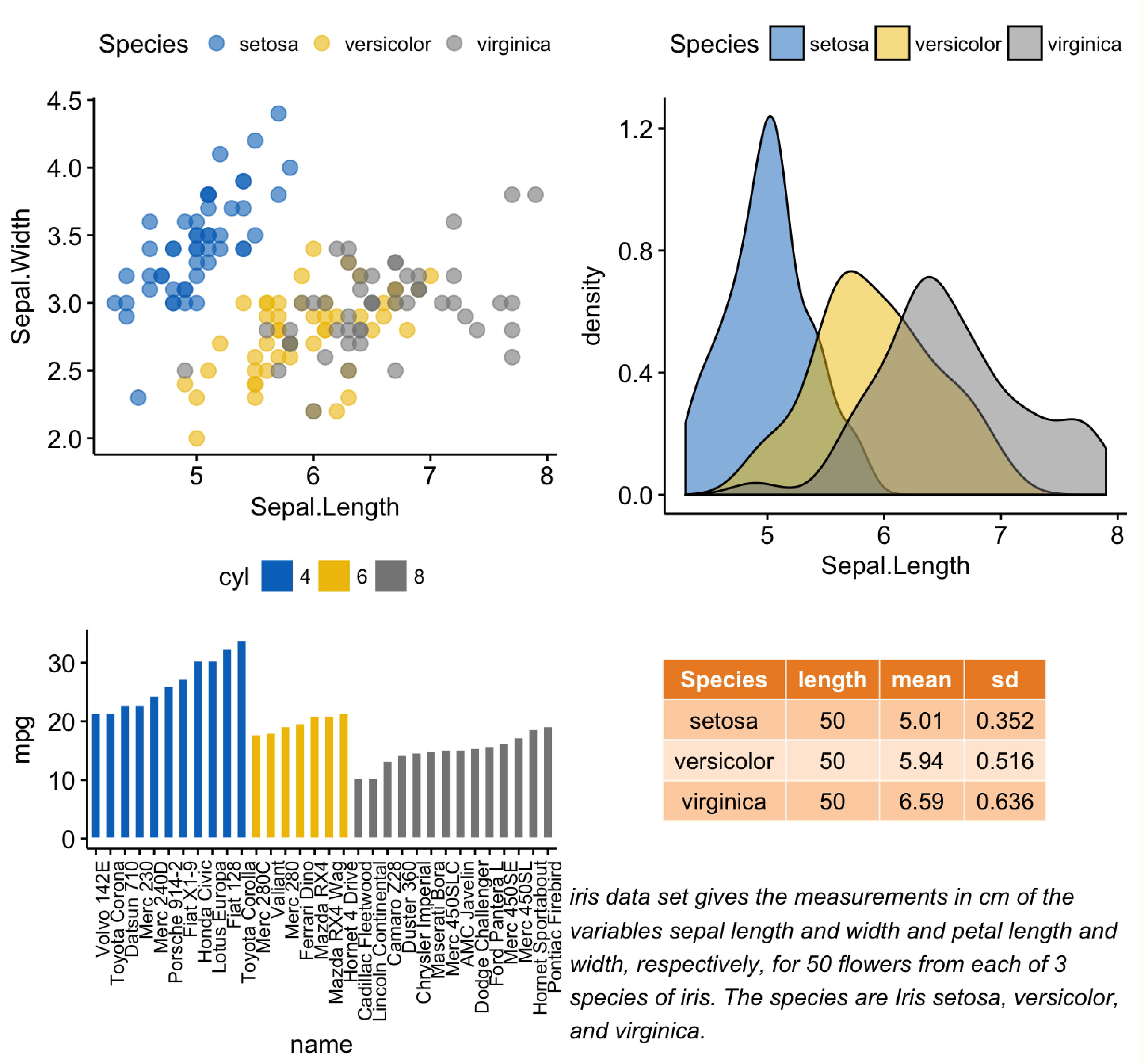
October 23, 2025

Recipe of the Day!

Harry Potter Sorting Hat Cookies

Announcements

- Homework #3 due Friday, October 24 by 11:59pm
- No lab this week
- Midterm will be released on **Monday, October 27** and due **Friday, November 7**
    - No late days may be used



[Combining plots and tables]

Coding question of the day

Using the **`gapminder`** dataset, create a categorical life expectancy variable and add it to the **`gapminder`** dataset. Call this variable **`life_expectancy_category`** and use the **`case_when`** function to create it. Life expectancy below 50 should be labeled "Low", life expectancy greater than or equal to 50 and less than 75 should be labeled "Medium", and life expectancy greater than or equal to 75 should be labeled "High".

Now, filter the dataset to only include observations from the year 1999, and create a bar chart of the 3 life expectancy categories. Reorder the bars so that they are in the order Low, Medium, High.

Be sure to run this code first

```
library(dslabs)
library(dplyr)
library(ggplot2)
data(gapminder)
```

Summarizing data

Summarizing Data Roadmap

1. Numerical summaries
2. Frequency tables
3. The `tableone` package
4. Reporting missing data

Summarizing data

Numerical summaries

Heather Mattie, PhD

Numerical summaries

Numerical summaries in statistics are concise ways to describe the key characteristics of a dataset using numbers

Measures of centrality

- Mean, median

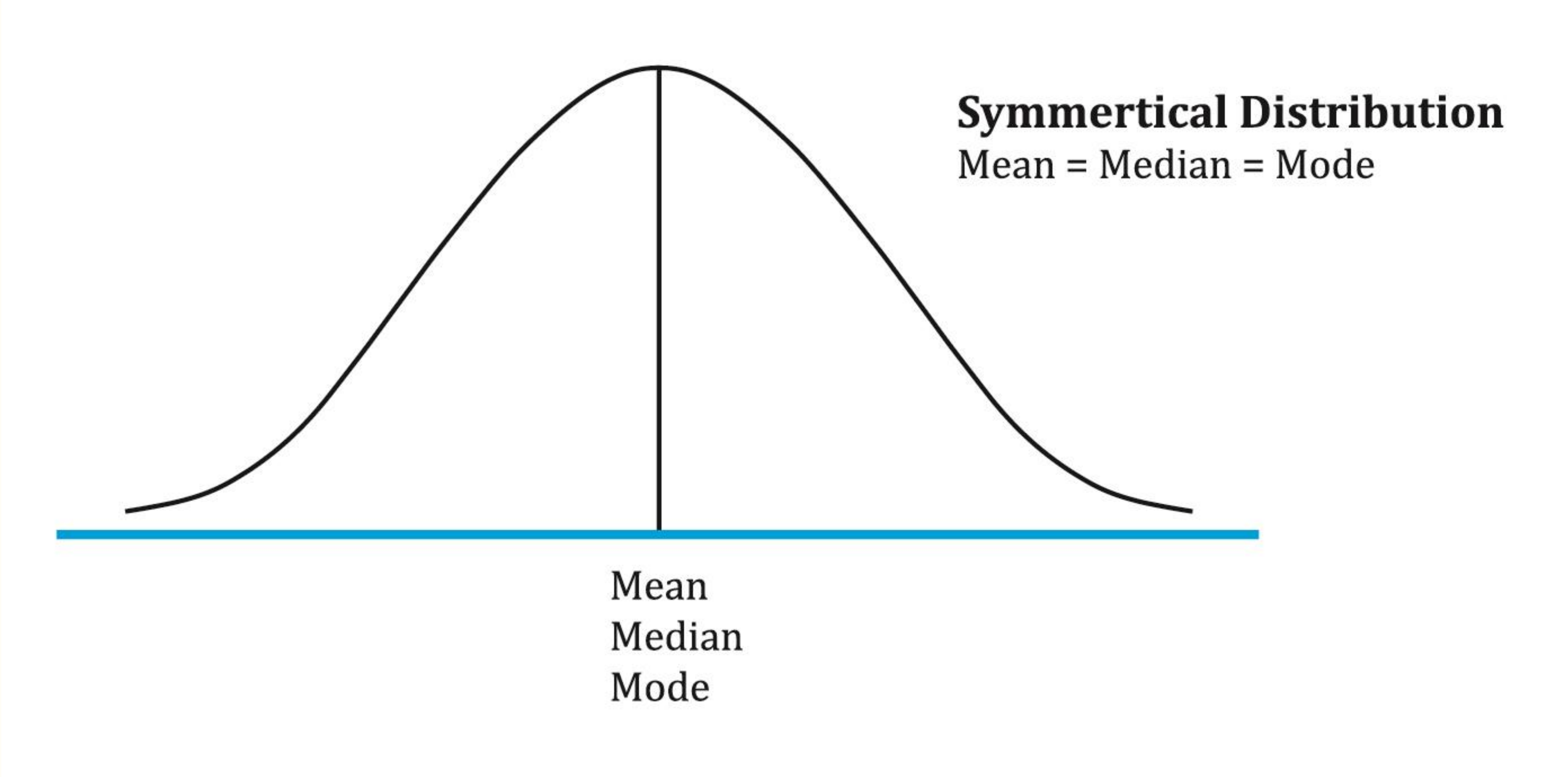
Measures of dispersion

- Standard deviation, variance, range, interquartile range

Mean

The mean is the average of all values in a dataset, calculated by summing all values and dividing by the total number of values.

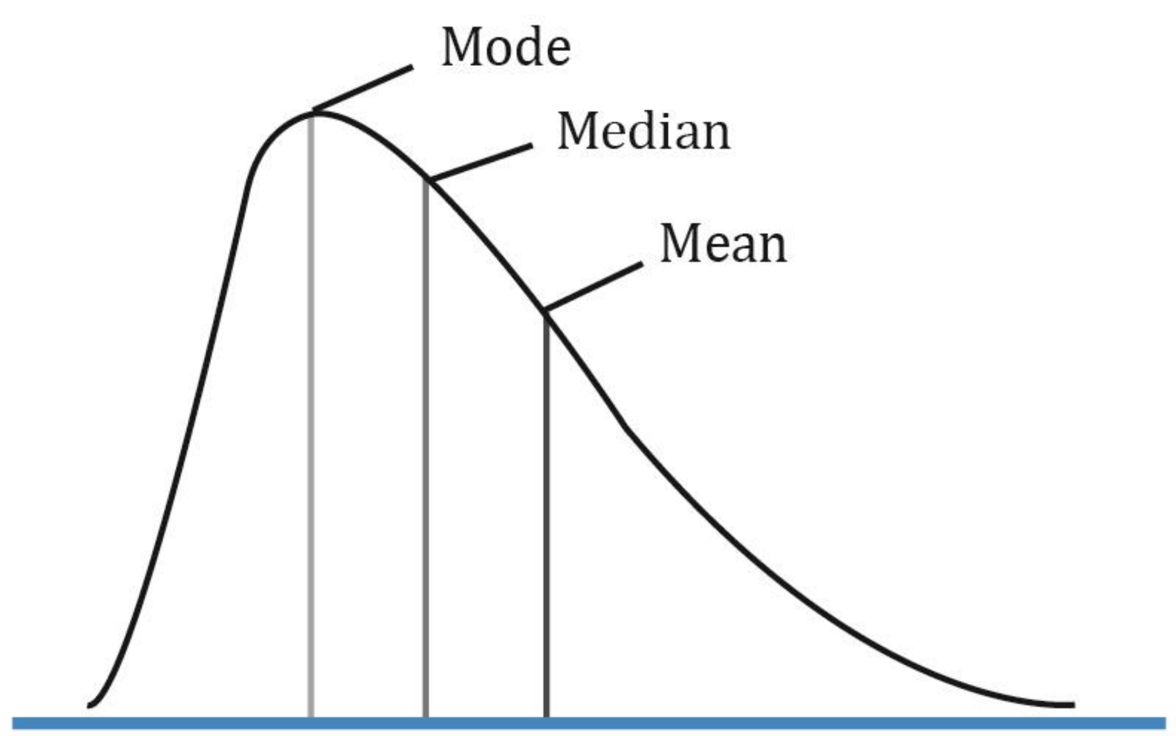
- The mean is the measure of centrality reported for symmetric distributions of data



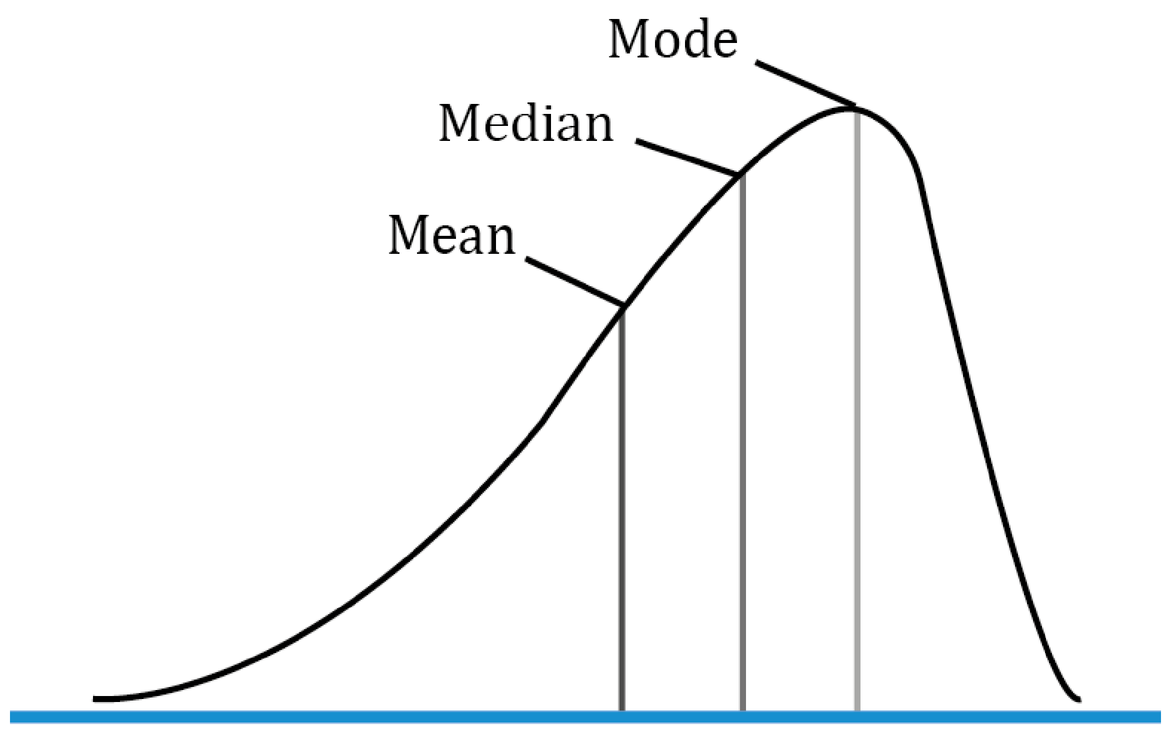
Median

The median is the middle value in a sorted dataset. It divides the data into two halves, with 50% of the values below it and 50% above it.

- The median is the measure of centrality reported for skewed distributions of data



Right - Skewed (Negative Skewness)



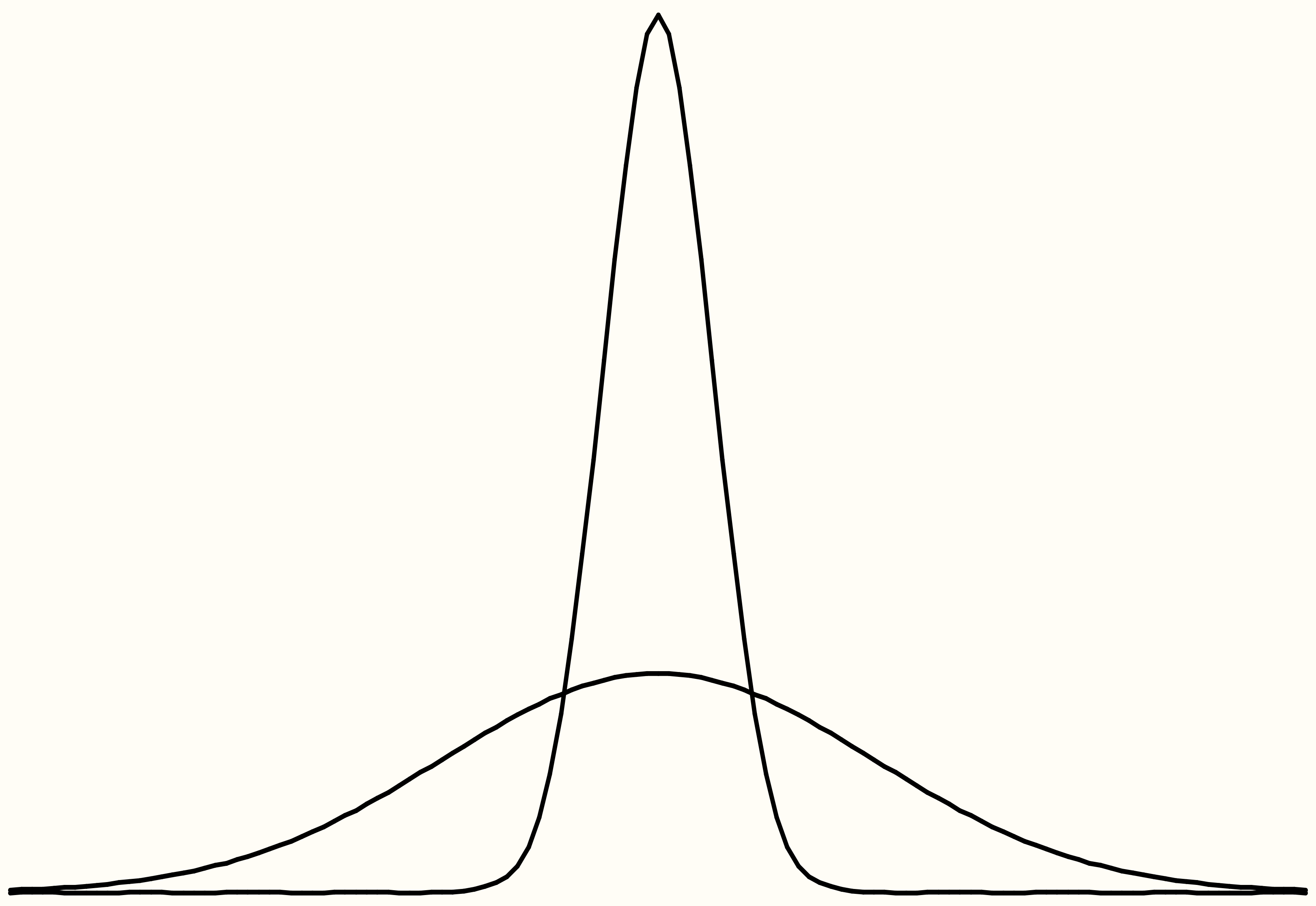
Left - Skewed (Negative Skewness)

Standard deviation and variance

Standard deviation is a measure of the spread of data around the mean. A higher standard deviation indicates greater variability.

- Standard deviation is the measure of dispersion reported for symmetric distributions of data

Variance is the square of standard deviation.

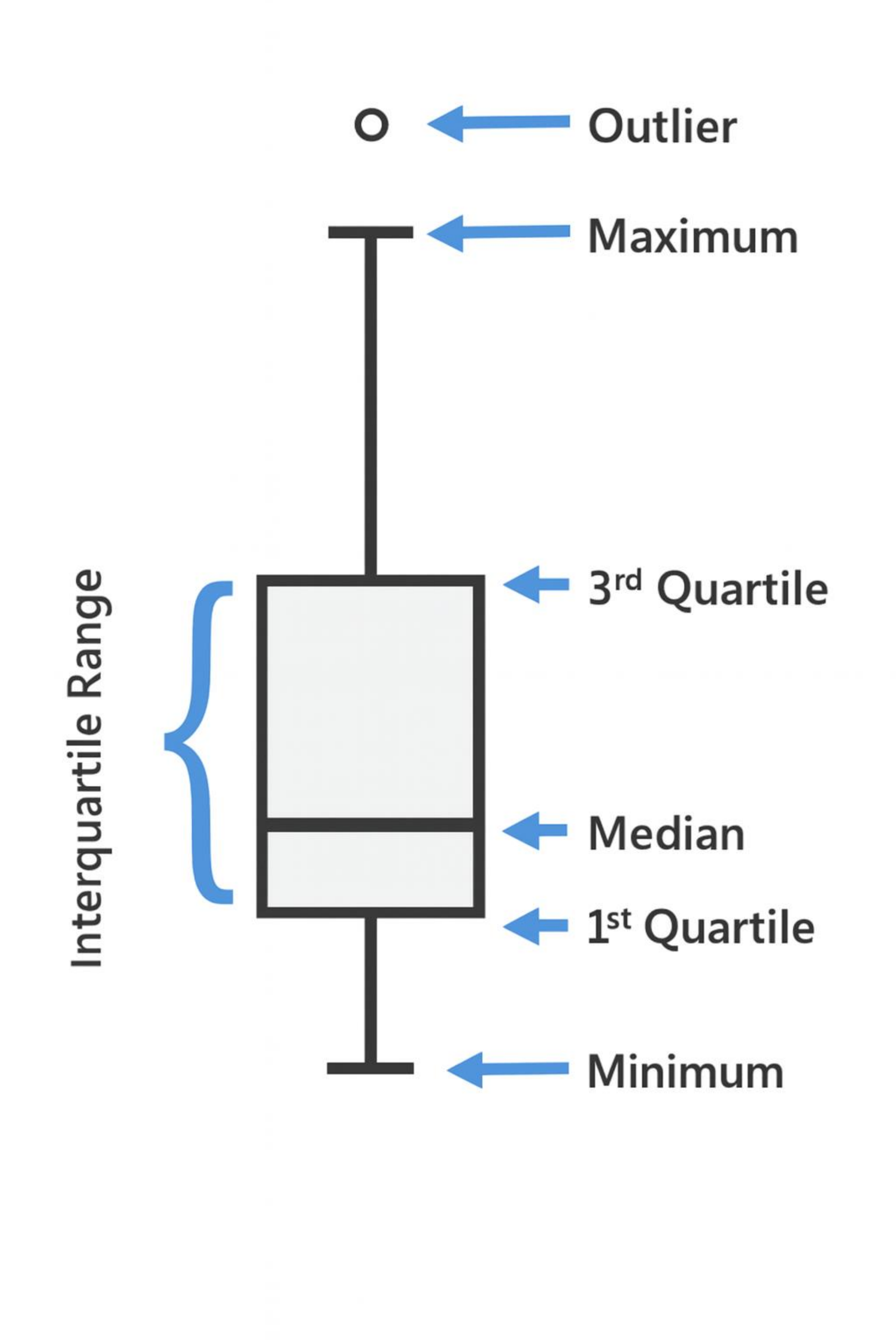


Range and interquartile range (IQR)

The range is the difference between the largest and smallest values in a dataset.

The interquartile range (IQR) is the difference between the third quartile (75th percentile) and the first quartile (25th percentile), representing the spread of the middle 50% of the data.

- The range or IQR is the measure of spread reported for skewed distributions



Numerical summaries in R

The **summary** function will return the minimum value, first quartile, median, mean, third quartile, and maximum value

```r
library(dslabs)
library(tidyverse)
data(gapminder)
summary(gapminder)
```

```
##                  country           year      infant_mortality life_expectancy
##  Albania              :   57   Min.   :1960   Min.   :  1.50   Min.   :13.20  
##  Algeria              :   57   1st Qu.:1974   1st Qu.: 16.00   1st Qu.:57.50  
##  Angola               :   57   Median :1988   Median : 41.50   Median :67.54  
##  Antigua and Barbuda:   57   Mean   :1988   Mean   : 55.31   Mean   :64.81  
##  Argentina            :   57   3rd Qu.:2002   3rd Qu.: 85.10   3rd Qu.:73.00  
##  Armenia              :   57   Max.   :2016   Max.   :276.90   Max.   :83.90  
##  (Other)              :10203                  NA's   :1453                    
##    fertility        population             gdp               continent   
##  Min.   :0.840   Min.   :3.124e+04   Min.   :4.040e+07   Africa  :2907  
##  1st Qu.:2.200   1st Qu.:1.333e+06   1st Qu.:1.846e+09   Americas:2052  
##  Median :3.750   Median :5.009e+06   Median :7.794e+09   Asia    :2679  
##  Mean   :4.084   Mean   :2.701e+07   Mean   :1.480e+11   Europe  :2223  
##  3rd Qu.:6.000   3rd Qu.:1.523e+07   3rd Qu.:5.540e+10   Oceania : 684  
##  Max.   :9.220   Max.   :1.376e+09   Max.   :1.174e+13                  
##  NA's   :187     NA's   :185         NA's   :2972                       
##              region    
##  Western Asia   :1026  
##  Eastern Africa : 912  
##  Western Africa : 912  
##  Caribbean      : 741  
##  South America  : 684  
##  Southern Europe: 684  
##  (Other)        :5586  
```

Numerical summaries in R

The **`summary`** function will return the minimum value, first quartile, median, mean, third quartile, and maximum value

```
gapminder %>%
  filter(year == 1986) %>%
  select(life_expectancy) %>%
  summary()
```

```
##  life_expectancy
##  Min.   :41.39
##  1st Qu.:58.96
##  Median :68.48
##  Mean   :65.57
##  3rd Qu.:71.85
##  Max.   :78.38
```

Numerical summaries in R

The **`mean`** function returns the mean

The **`median`** function returns the median

```
gapminder_1986 <- gapminder %>%
  filter(year == 1986 & !is.na(life_expectancy))

le_1986 <- gapminder_1986$life_expectancy

mean(le_1986)
```

```
## [1] 65.57362
```

```
median(le_1986)
```

```
## [1] 68.48
```

Numerical summaries in R

The `sd` function returns the standard deviation

The `var` function returns the variance

The `range` function returns the minimum and maximum values

The `IQR` function returns the interquartile range

```
sd(le_1986)
```

```
## [1] 8.679824
```

```
var(le_1986)
```

```
## [1] 75.33935
```

```
range(le_1986)
```

```
## [1] 41.39 78.38
```

```
IQR(le_1986)
```

```
## [1] 12.89
```

Numerical summaries in R

The **group_by** and **summarize** functions return summary values for different groups

- Always **ungroup()** when finished using **group_by**

```
gapminder %>%
  filter(year == 1986 & !is.na(life_expectancy)) %>%
  group_by(continent) %>%
  summarize(le_mean = mean(life_expectancy),
            le_median = median(life_expectancy),
            le_sd = sd(life_expectancy),
            le_var = var(life_expectancy),
            le_range = range(life_expectancy),
            le_IQR = IQR(life_expectancy)) %>%
  ungroup()
```

```
## # A tibble: 10 × 7
##    continent le_mean le_median le_sd le_var le_range le_IQR
##    <fct>       <dbl>     <dbl> <dbl>  <dbl>    <dbl>  <dbl>
##  1 Africa       56.0      55.5  7.03  49.5      41.4   9.82
##  2 Africa       56.0      55.5  7.03  49.5      71.6   9.82
##  3 Americas     69.4      69.8  4.97  24.7      52.8   4.78
##  4 Americas     69.4      69.8  4.97  24.7      76.5   4.78
##  5 Asia         66.5      68.3  6.84  46.8      49.0   9.40
##  6 Asia         66.5      68.3  6.84  46.8      78.2   9.40
##  7 Europe       73.5      74.1  2.63   6.93     67.3   4.57
##  8 Europe       73.5      74.1  2.63   6.93     78.4   4.57
##  9 Oceania      65.2      65.4  6.52  42.6      55.2   8.54
## 10 Oceania      65.2      65.4  6.52  42.6      76.1   8.54
```

Numerical summaries in R

The `group_by` and `reframe` functions return summary values for different groups

- Do not need to include `ungroup()` when using the reframe function

```
gapminder %>%
  filter(year == 1986 & !is.na(life_expectancy)) %>%
  group_by(continent) %>%
  reframe(le_mean = mean(life_expectancy),
            le_median = median(life_expectancy),
            le_sd = sd(life_expectancy),
            le_var = var(life_expectancy),
            le_range = range(life_expectancy),
            le_IQR = IQR(life_expectancy)) #%>%
```

```
## # A tibble: 10 × 7
##    continent le_mean le_median le_sd le_var le_range le_IQR
##    <fct>       <dbl>     <dbl> <dbl>  <dbl>    <dbl>  <dbl>
##  1 Africa       56.0      55.5  7.03  49.5      41.4   9.82
##  2 Africa       56.0      55.5  7.03  49.5      71.6   9.82
##  3 Americas     69.4      69.8  4.97  24.7      52.8   4.78
##  4 Americas     69.4      69.8  4.97  24.7      76.5   4.78
##  5 Asia         66.5      68.3  6.84  46.8      49.0   9.40
##  6 Asia         66.5      68.3  6.84  46.8      78.2   9.40
##  7 Europe       73.5      74.1  2.63   6.93     67.3   4.57
##  8 Europe       73.5      74.1  2.63   6.93     78.4   4.57
##  9 Oceania      65.2      65.4  6.52  42.6      55.2   8.54
## 10 Oceania      65.2      65.4  6.52  42.6      76.1   8.54
```

Summary

- Numerical summaries are concise ways to describe the key characteristics of a dataset using numbers
- The mean and standard deviation are reported when the data distribution is symmetric
- The median, range, and interquartile range are reported when the data distribution is skewed
- The are several numerical summary functions available in base R and the `tidyverse` package

Summarizing data

Frequency tables

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Frequency tables

- Frequency tables are used to summarize nominal and ordinal measurements falling into categories
- They list each of the possible values that the data can assume (the categories or classes), along with the number of times each value occurs
- Categories must not overlap, and are usually (but not always) chosen to be of equal width

Frequency tables

The `table` function displays a table of each unique value and its corresponding count

```
library(dslabs)
library(tidyverse)
data(gapminder)

gapminder <- gapminder %>%
  filter(year == 2000 & !is.na(fertility)) %>%
  mutate(fertility_cat = round(fertility))

fertility_table <- table(gapminder$continent, gapminder$fertility_cat)
fertility_table
```

```
##
##             1  2  3  4  5  6  7  8
##   Africa    0  3  6  6 10 17  8  1
##   Americas  0 17 13  5  1  0  0  0
##   Asia      6 14 12 11  2  1  1  0
##   Europe   22 17  0  0  0  0  0  0
##   Oceania   0  4  1  5  2  0  0  0
```

Frequency tables

The **`table`** function displays a table of each unique value and its corresponding count

- Using the **`with`** function to indicate which data frame to use can cut down on code

```
fertility_table <- with(gapminder, table(continent, fertility_cat))
fertility_table
```

```
##           fertility_cat
## continent   1  2  3  4  5  6  7  8
##   Africa    0  3  6  6 10 17  8  1
##   Americas  0 17 13  5  1  0  0  0
##   Asia      6 14 12 11  2  1  1  0
##   Europe   22 17  0  0  0  0  0  0
##   Oceania   0  4  1  5  2  0  0  0
```

Relative frequency tables

- Relative frequency is the proportion or percent of the whole that falls into a given category
- Divide the number of subjects within a category by the total number in the table for a proportion; multiply the proportion by 100 for a percent
- Relative frequencies can be used to compare populations of different sizes

Relative frequency tables

The **prop.table** function returns a relative frequency table

```
fertility_prop_table <- prop.table(fertility_table)
fertility_prop_table
```

```
##           fertility_cat
## continent            1           2           3           4           5
##   Africa    0.000000000 0.016216216 0.032432432 0.032432432 0.054054054
##   Americas  0.000000000 0.091891892 0.070270270 0.027027027 0.005405405
##   Asia      0.032432432 0.075675676 0.064864865 0.059459459 0.010810811
##   Europe    0.118918919 0.091891892 0.000000000 0.000000000 0.000000000
##   Oceania   0.000000000 0.021621622 0.005405405 0.027027027 0.010810811
##           fertility_cat
## continent            6           7           8
##   Africa    0.091891892 0.043243243 0.005405405
##   Americas  0.000000000 0.000000000 0.000000000
##   Asia      0.005405405 0.005405405 0.000000000
##   Europe    0.000000000 0.000000000 0.000000000
##   Oceania   0.000000000 0.000000000 0.000000000
```

Relative frequency tables

The **`prop.table`** function returns a relative frequency table

- Using the **`round`** function to limit the number of decimal places provided in the table can make the output easier to read and interpret

```
fertility_prop_table <- round(prop.table(fertility_table), 2)
fertility_prop_table
```

```
##           fertility_cat
## continent     1    2    3    4    5    6    7    8
##   Africa   0.00 0.02 0.03 0.03 0.05 0.09 0.04 0.01
##   Americas 0.00 0.09 0.07 0.03 0.01 0.00 0.00 0.00
##   Asia     0.03 0.08 0.06 0.06 0.01 0.01 0.01 0.00
##   Europe   0.12 0.09 0.00 0.00 0.00 0.00 0.00 0.00
##   Oceania  0.00 0.02 0.01 0.03 0.01 0.00 0.00 0.00
```

Relative frequency tables

The **proportions** function also returns a relative frequency table

```
fertility_prop_table <- round(proportions(fertility_table), 2)
fertility_prop_table
```

```
##           fertility_cat
## continent     1    2    3    4    5    6    7    8
##   Africa   0.00 0.02 0.03 0.03 0.05 0.09 0.04 0.01
##   Americas 0.00 0.09 0.07 0.03 0.01 0.00 0.00 0.00
##   Asia     0.03 0.08 0.06 0.06 0.01 0.01 0.01 0.00
##   Europe   0.12 0.09 0.00 0.00 0.00 0.00 0.00 0.00
##   Oceania  0.00 0.02 0.01 0.03 0.01 0.00 0.00 0.00
```

Relative frequency tables

The `proportions` function also returns a relative frequency table

- Multiplying but 100 returns percentages in the table

```
fertility_prop_table <- (round(proportions(fertility_table), 2) * 100)
fertility_prop_table
```

```
##           fertility_cat
## continent   1  2  3  4  5  6  7  8
##   Africa    0  2  3  3  5  9  4  1
##   Americas  0  9  7  3  1  0  0  0
##   Asia      3  8  6  6  1  1  1  0
##   Europe   12  9  0  0  0  0  0  0
##   Oceania   0  2  1  3  1  0  0  0
```

Summary

- Frequency tables summarize categorical data by displaying the number of times each value occurs
- Relative frequency tables summarize categorical data by displaying the proportion of values that fall into a given category
- The values in a relative frequency table can be transformed into percentages by multiplying each value by 100

Summarizing data

The `tableone` package

Heather Mattie, PhD

HARVARD T.H. CHAN
SCHOOL OF PUBLIC HEALTH

Organizing data summaries

- Oftentimes, it is useful to have numerical summaries, frequency table values, and relative frequency table values presented in one table
- This table is usually the first table in a scientific journal article and is referred to as "Table 1"

Table 1. Participant Demographics and Characteristics (N=504)

Characteristic	**Total, *n* (%) N=504**	**Noticed calories, *n* (%) *n*=277 (55%)**	**Did not notice calories, *n* (%) *n*=227 (45%)**	**Chi-square or Wilcoxon, *p*-value**
Participant age, in years				
18–25	117 (23.2%)	56 (20.2%)	61 (26.9%)	2.738, 0.0980
26–39	168 (33.3%)	96 (34.7%)	72 (31.7%)	0.362, 0.5476
≥40	219 (43.5%)	125 (45.1%)	94 (41.4%)	0.558, 0.4550
Participant race				
White	358 (71.0%)	207 (74.7%)	151 (66.5%)	3.697, 0.0545
Asian/Asian American	28 (5.6%)	15 (5.4%)	13 (5.7%)	<0.001, 0.9999
Black or African American	72 (14.3%)	33 (11.9%)	39 (17.2%)	2.413, 0.1203

tableone

The **tableone** package eases the creation of a “Table 1”

- Created in 2022 by a Harvard T.H. Chan alumnus
- Supports continuous and categorical variables
- Can provide p-values
- Supports stratification by groups

Package ‘tableone’

July 22, 2025

Type Package

Title Create 'Table 1' to Describe Baseline Characteristics with or without Propensity Score Weights

Version 0.13.2

Date 2022-04-15

Maintainer Kazuki Yoshida <kazukiyoshida@mail.harvard.edu>

Description Creates 'Table 1', i.e., description of baseline patient characteristics, which is essential in every medical research. Supports both continuous and categorical variables, as well as p-values and standardized mean differences. Weighted data are supported via the 'survey' package.

License GPL-2

Imports survey, MASS, e1071, zoo, gmodels, nlme, labelled

Suggests survival, testthat, Matrix, Matching, reshape2, ggplot2, knitr, geepack, lme4, lmerTest, rmarkdown

URL https://github.com/kaz-yos/tableone

tableone

```r
library(dslabs)
library(tidyverse)
library(tableone)
data(gapminder)

table_df <- gapminder %>%
  filter(year == 1990)
table_df <- table_df[complete.cases(table_df),]

t1 <- CreateTableOne(vars = names(table_df), data = table_df)
print(t1)
```

```
##
##                                            Overall
##   n                                           164
##   country (%)
##      Albania                                    1 ( 0.6)
##      Algeria                                    1 ( 0.6)
##      Angola                                     1 ( 0.6)
##      Antigua and Barbuda                        1 ( 0.6)
##      Argentina                                  1 ( 0.6)
##      Armenia                                    1 ( 0.6)
##      Aruba                                      0 ( 0.0)
##      Australia                                  1 ( 0.6)
##      Austria                                    1 ( 0.6)
##      Azerbaijan                                 1 ( 0.6)
##      Bahamas                                    1 ( 0.6)
##      Bahrain                                    1 ( 0.6)
##      Bangladesh                                 1 ( 0.6)
##      Barbados                                   1 ( 0.6)
```

tableone

```r
library(dslabs)
library(tidyverse)
library(tableone)
data(gapminder)

table_df <- gapminder %>%
  filter(year == 1990)
table_df <- table_df[complete.cases(table_df),]

t1 <- CreateTableOne(vars = names(table_df), data = table_df)
print(t1)
```

We don't need to print the **country**, **year**, or **region** variables

```
##      Zambia                                   1 ( 0.6)
##      Zimbabwe                                 1 ( 0.6)
##   year (mean (SD))                      1990.00 (0.00)
##   infant_mortality (mean (SD))            49.70 (40.50)
##   life_expectancy (mean (SD))             66.32 (8.56)
##   fertility (mean (SD))                    3.97 (1.89)
##   population (mean (SD))            31297074.62 (115897913.10)
##   gdp (mean (SD))                 145422849909.40 (664869857190.42)
##   continent (%)
##      Africa                                  49 (29.9)
##      Americas                                32 (19.5)
##      Asia                                    38 (23.2)
##      Europe                                  35 (21.3)
##      Oceania                                 10 ( 6.1)
##   region (%)
##      Australia and New Zealand                2 ( 1.2)
##      Caribbean                               10 ( 6.1)
##      Central America                          8 ( 4.9)
##      Central Asia                             5 ( 3.0)
##      Eastern Africa                          15 ( 9.1)
##      Eastern Asia                             4 ( 2.4)
##      Eastern Europe                          10 ( 6.1)
##      Melanesia                                4 ( 2.4)
##      Micronesia                               2 ( 1.2)
##      Middle Africa                            8 ( 4.9)
##      Northern Africa                          5 ( 3.0)
##      Northern America                         2 ( 1.2)
```

tableone

```r
vars <- c("infant_mortality", "life_expectancy",
          "fertility", "population", "gdp",
          "continent")

t1 <- CreateTableOne(vars = vars, data = table_df)
options(digits = 2)
kableone(t1)
```

	Overall
n	164
infant_mortality (mean (SD))	49.70 (40.50)
life_expectancy (mean (SD))	66.32 (8.56)
fertility (mean (SD))	3.97 (1.89)
population (mean (SD))	31297074.62 (115897913.10)
gdp (mean (SD))	145422849909.40 (664869857190.42)
continent (%)	
Africa	49 (29.9)
Americas	32 (19.5)
Asia	38 (23.2)
Europe	35 (21.3)
Oceania	10 (6.1)

Variable distributions

The default for the `CreateTableOne` function is to report the mean and standard deviation for each continuous (numeric) variable

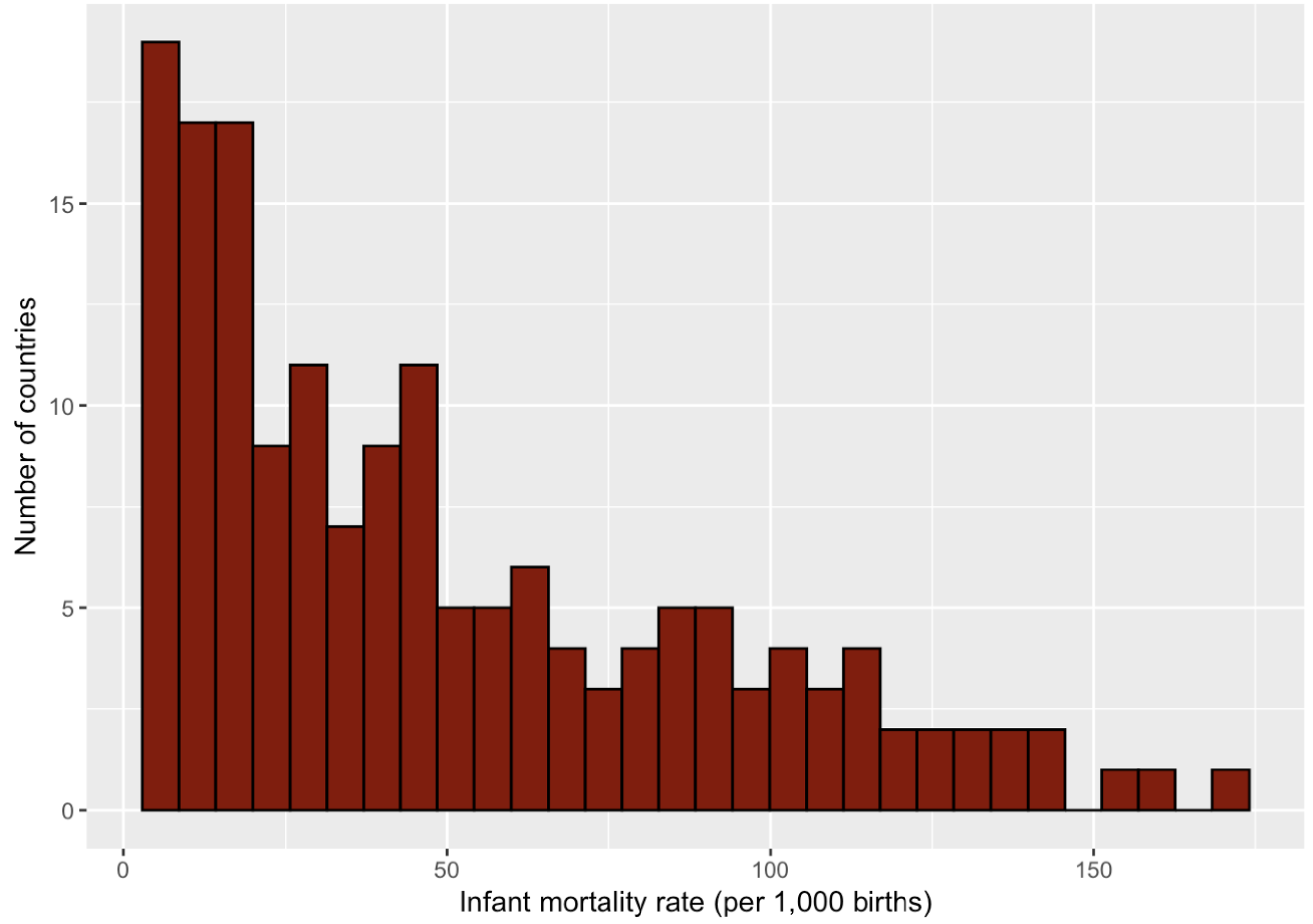
- Need to check that the distribution of each continuous variable is normally distributed – if not, need to report the median and IQR

Variable distributions

The default for the **CreateTableOne** function is to report the mean and standard deviation for each continuous (numeric) variable

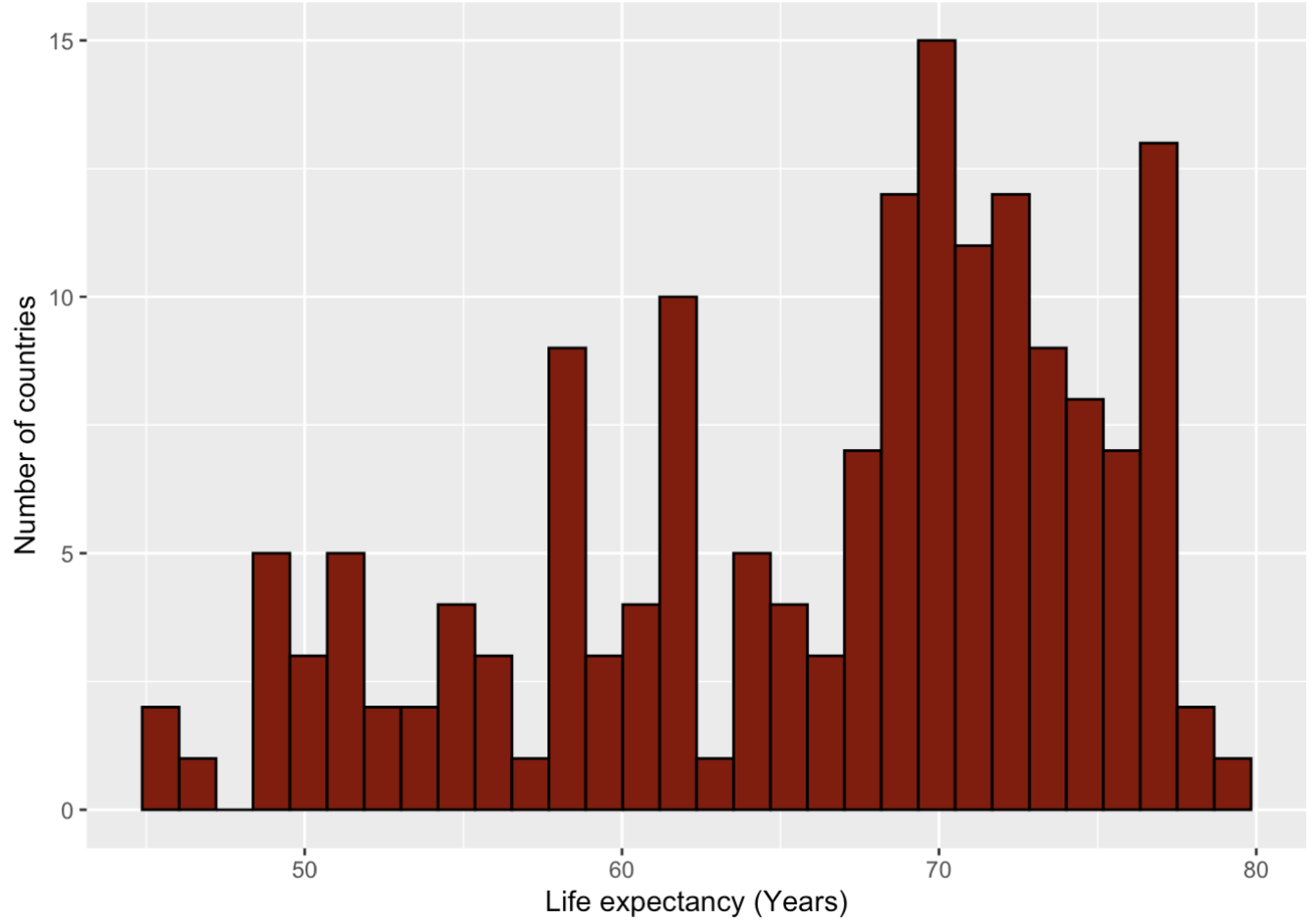
- Need to check that the distribution of each continuous variable is normally distributed – if not, need to report the median and IQR

```
table_df %>%
  ggplot(aes(infant_mortality)) +
  geom_histogram(color = "black", fill = "darkred") +
  xlab("Infant mortality rate (per 1,000 births)") +
  ylab("Number of countries")
```

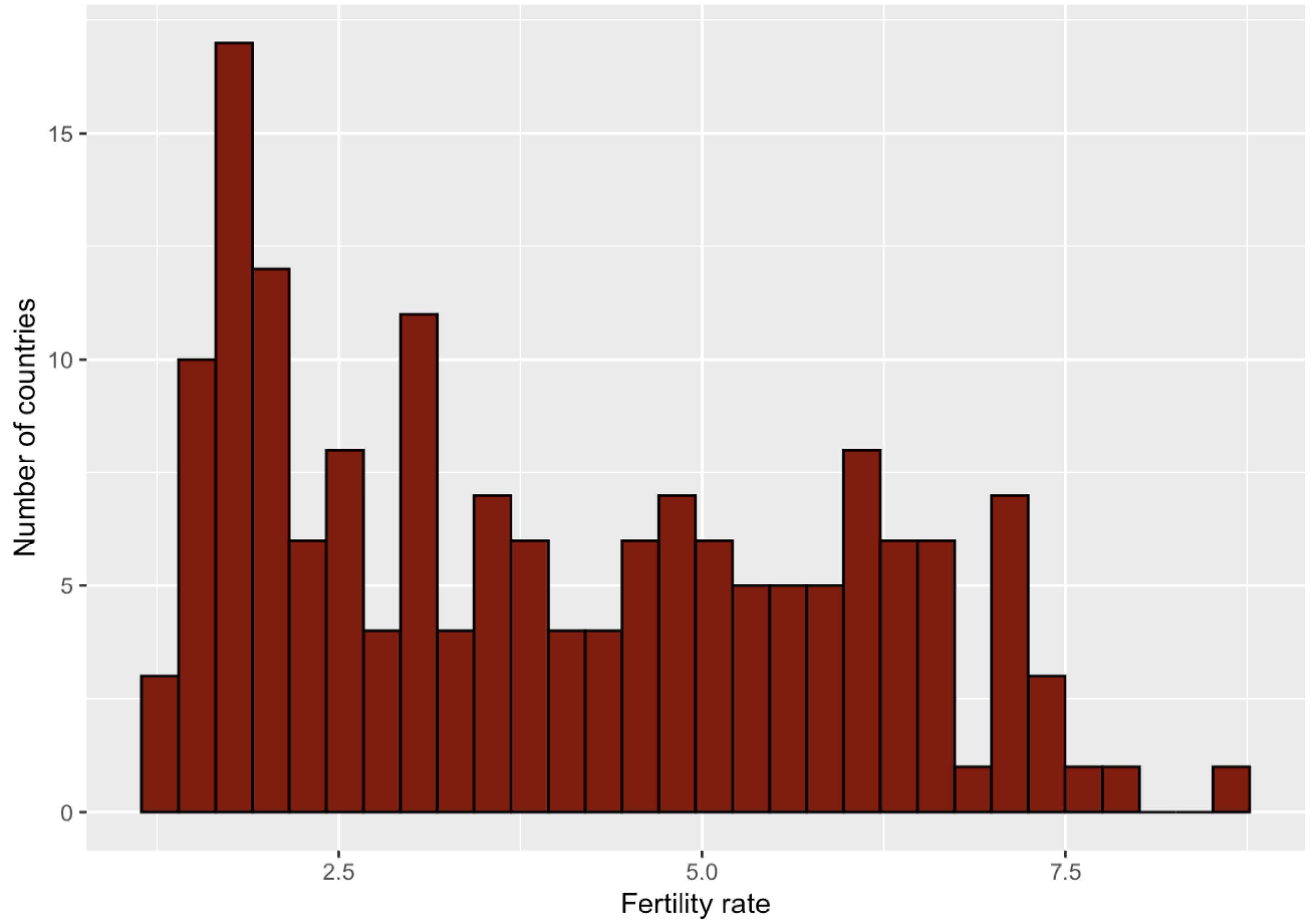


Variable distributions

```r
table_df %>%
  ggplot(aes(life_expectancy)) +
  geom_histogram(color = "black", fill = "darkred") +
  xlab("Life expectancy (Years)") +
  ylab("Number of countries")
```

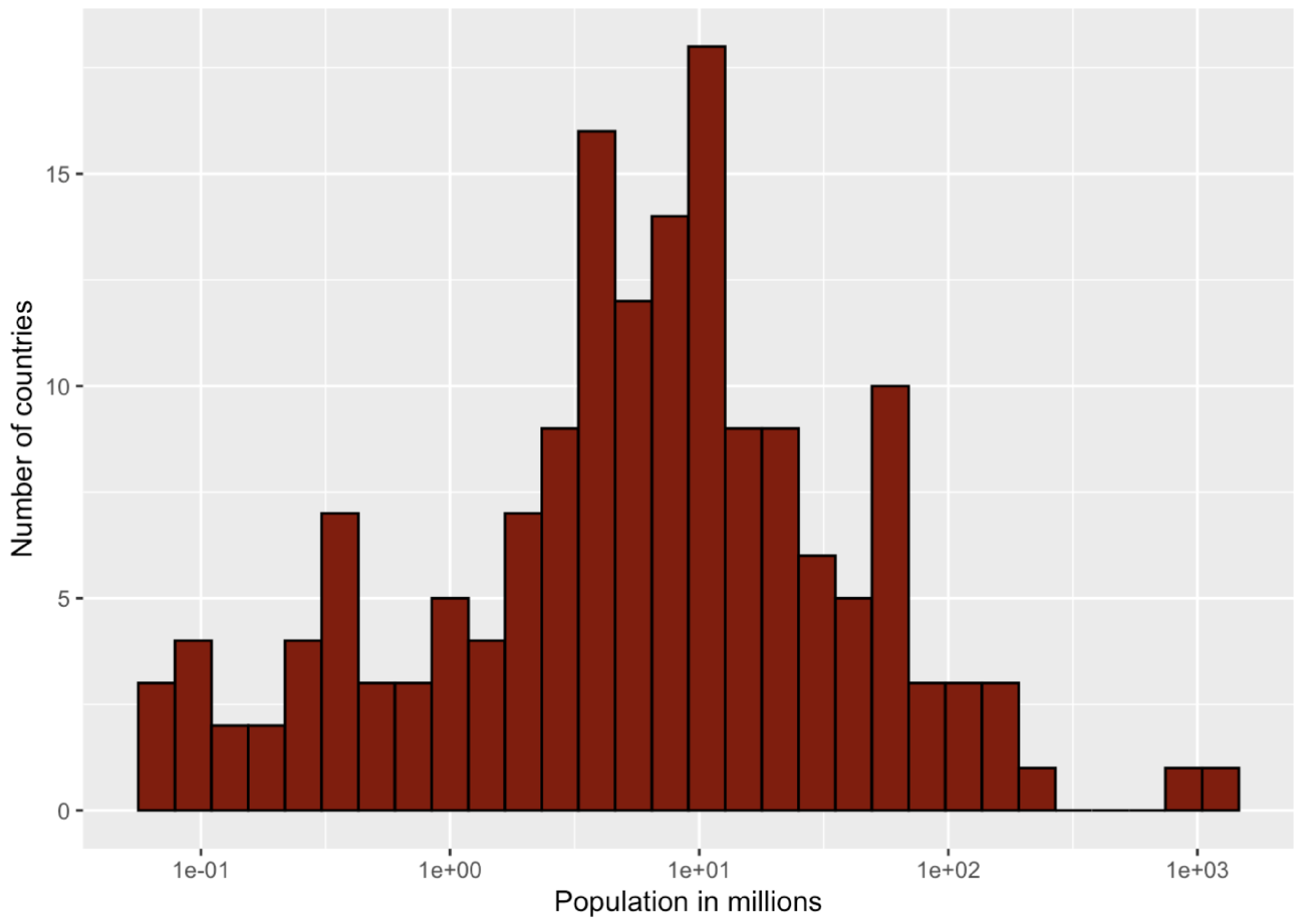



```r
table_df %>%
  ggplot(aes(fertility)) +
  geom_histogram(color = "black", fill = "darkred") +
  xlab("Fertility rate") +
  ylab("Number of countries")
```

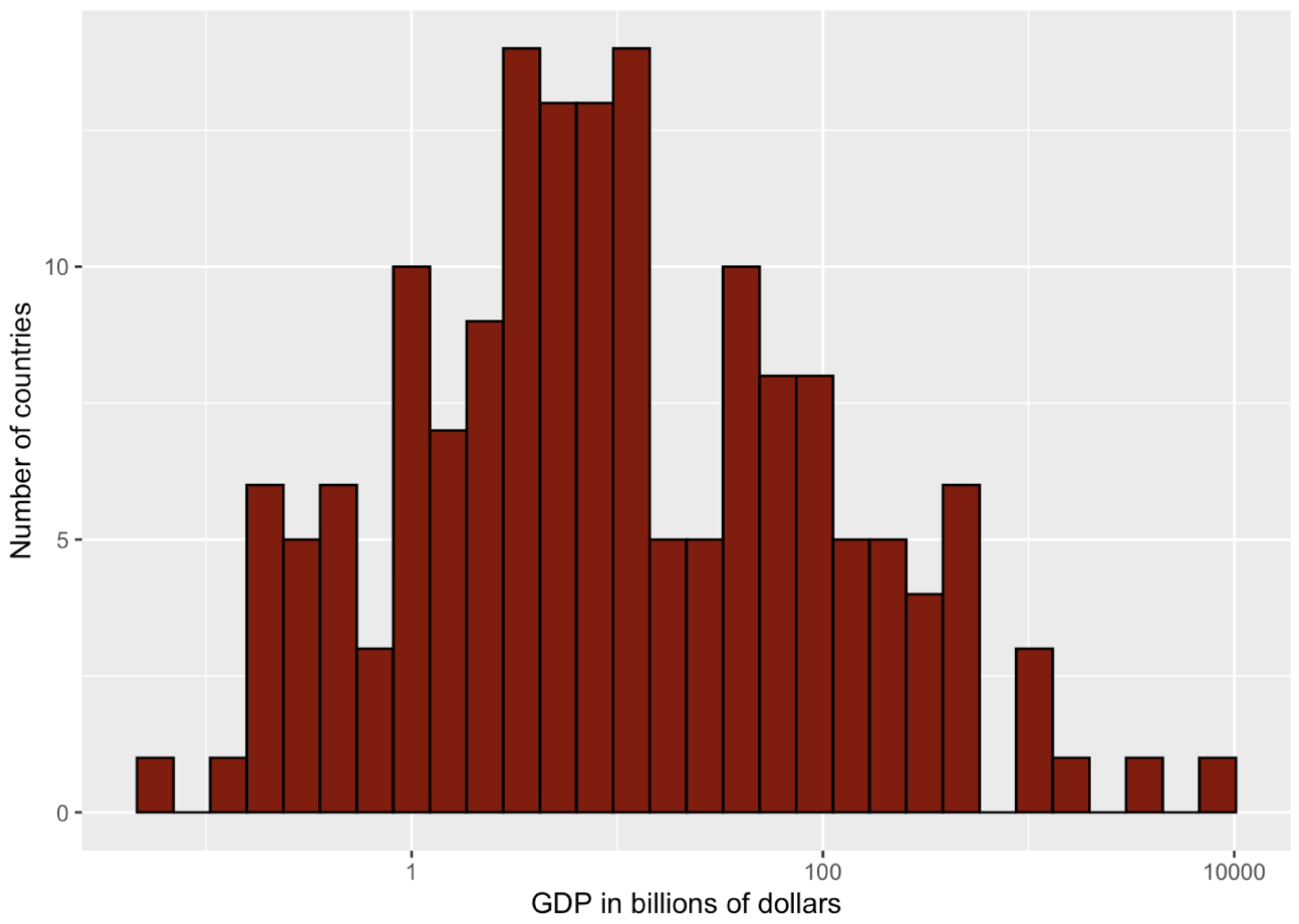


Variable distributions

```
table_df %>%
  ggplot(aes(population/10^6)) +
  geom_histogram(color = "black", fill = "darkred") +
  xlab("Population in millions") +
  ylab("Number of countries") +
  scale_x_log10()
```




```
table_df %>%
  ggplot(aes(gdp/10^9)) +
  geom_histogram(color = "black", fill = "darkred") +
  xlab("GDP in billions of dollars") +
  ylab("Number of countries") +
  scale_x_log10()
```



nonormal

Define a vector of column names containing all variables that are not normally distributed

- Use the argument **nonnormal =** and the vector created in the **kableone** function

```
nonNormalVars <- c("infant_mortality", "life_expectancy",
                   "fertility", "population", "gdp")

t1 <- CreateTableOne(vars = vars, data = table_df)
options(digits = 2)
kableone(t1, nonnormal = nonNormalVars)
```

	Overall
n	164
infant_mortality (median [IQR])	38.40 [16.00, 77.15]
life_expectancy (median [IQR])	69.15 [60.40, 72.53]
fertility (median [IQR])	3.70 [2.16, 5.59]
population (median [IQR])	6765083.00 [2019378.75, 18343744.75]
gdp (median [IQR])	8224534133.00 [2170758367.50, 51772082085.75]
continent (%)	
Africa	49 (29.9)
Americas	32 (19.5)
Asia	38 (23.2)
Europe	35 (21.3)
Oceania	10 (6.1)

strata

Stratification by a categorical variable is supported

- Use the **strata =** argument and include the categorical variable name in quotes

```
vars <- c("infant_mortality", "life_expectancy",
          "fertility", "population", "gdp")

nonNormalVars <- c("infant_mortality", "life_expectancy",
                   "fertility", "population", "gdp")

t1 <- CreateTableOne(vars = vars, data = table_df,
                     strata = "continent")
options(digits = 2)
kableone(t1, nonnormal = nonNormalVars)
```

	Africa	Americas	Asia	Europe	Oceania	p	test
n	49	32	38	35	10		
infant_mortality (median [IQR])	92.70 [63.00, 119.80]	26.30 [19.45, 45.30]	41.45 [21.35, 74.75]	11.20 [7.35, 16.65]	27.50 [20.22, 40.60]	<0.001	nonnorm
life_expectancy (median [IQR])	56.10 [51.50, 61.30]	71.50 [69.55, 73.35]	68.70 [64.33, 71.72]	74.20 [71.20, 76.75]	64.70 [60.55, 69.10]	<0.001	nonnorm
fertility (median [IQR])	6.21 [5.35, 6.65]	3.09 [2.60, 3.83]	3.80 [2.87, 5.08]	1.81 [1.57, 2.00]	4.72 [3.71, 4.95]	<0.001	nonnorm
population (median [IQR])	7514201.00 [1597534.00, 13371971.00]	4558551.50 [642280.50, 14821391.25]	12206319.00 [3405013.50, 55625548.25]	8821111.00 [4302244.50, 19202250.00]	237357.00 [108906.50, 2730307.00]	0.001	nonnorm
gdp (median [IQR])	2590637104.00 [1019207023.00, 7062040511.00]	7961843305.00 [2664939559.50, 37181495423.25]	13348138779.00 [4889136750.50, 69268340898.75]	71953291930.00 [14464231648.50, 193917649818.00]	267752123.70 [184648934.78, 2101457579.25]	<0.001	nonnorm

strata

Stratification by a categorical variable is supported

- Use the **strata =** argument and include the categorical variable name in quotes
- Can add an overall column back to the table with **addOverall = TRUE**

```r
vars <- c("infant_mortality", "life_expectancy",
          "fertility", "population", "gdp")

nonNormalVars <- c("infant_mortality", "life_expectancy",
                   "fertility", "population", "gdp")

t1 <- CreateTableOne(vars = vars, data = table_df,
                     strata = "continent", addOverall = TRUE)
options(digits = 2)
kableone(t1, nonnormal = nonNormalVars)
```

	Overall	Africa	Americas	Asia	Europe	Oceania	p	test
n	164	49	32	38	35	10		
infant_mortality (median [IQR])	38.40 [16.00, 77.15]	92.70 [63.00, 119.80]	26.30 [19.45, 45.30]	41.45 [21.35, 74.75]	11.20 [7.35, 16.65]	27.50 [20.22, 40.60]	<0.001	nonnorm
life_expectancy (median [IQR])	69.15 [60.40, 72.53]	56.10 [51.50, 61.30]	71.50 [69.55, 73.35]	68.70 [64.33, 71.72]	74.20 [71.20, 76.75]	64.70 [60.55, 69.10]	<0.001	nonnorm
fertility (median [IQR])	3.70 [2.16, 5.59]	6.21 [5.35, 6.65]	3.09 [2.60, 3.83]	3.80 [2.87, 5.08]	1.81 [1.57, 2.00]	4.72 [3.71, 4.95]	<0.001	nonnorm
population (median [IQR])	6765083.00 [2019378.75, 18343744.75]	7514201.00 [1597534.00, 13371971.00]	4558551.50 [642280.50, 14821391.25]	12206319.00 [3405013.50, 55625548.25]	8821111.00 [4302244.50, 19202250.00]	237357.00 [108906.50, 2730307.00]	0.001	nonnorm
gdp (median [IQR])	8224534133.00 [2170758367.50, 51772082085.75]	2590637104.00 [1019207023.00, 7062040511.00]	7961843305.00 [2664939559.50, 37181495423.25]	13348138779.00 [4889136750.50, 69268340898.75]	71953291930.00 [14464231648.50, 193917649818.00]	267752123.70 [184648934.78, 2101457579.25]	<0.001	nonnorm

Finishing touches

- Convert population and GDP values
- Rename variables

```r
table_df <- gapminder %>%
  filter(year == 1990) %>%
  mutate(Population = population/10^6,
         GDP = gdp/10^9) %>%
  rename("Infant mortality" = infant_mortality,
         "Life expectancy" = life_expectancy,
         "Fertility" = fertility)

table_df <- table_df[complete.cases(table_df),]

vars <- c("Infant mortality", "Life expectancy",
          "Fertility", "Population", "GDP")

nonNormalVars <- c("Infant mortality", "Life expectancy",
                   "Fertility", "Population", "GDP")

t1 <- CreateTableOne(vars = vars, data = table_df,
                     strata = "continent", addOverall = TRUE)
options(digits = 2)
kableone(t1, nonnormal = nonNormalVars)
```

Finishing touches

	Overall	Africa	Americas	Asia	Europe	Oceania	p	test
n	164	49	32	38	35	10		
Infant mortality (median [IQR])	38.40 [16.00, 77.15]	92.70 [63.00, 119.80]	26.30 [19.45, 45.30]	41.45 [21.35, 74.75]	11.20 [7.35, 16.65]	27.50 [20.22, 40.60]	<0.001	nonnorm
Life expectancy (median [IQR])	69.15 [60.40, 72.53]	56.10 [51.50, 61.30]	71.50 [69.55, 73.35]	68.70 [64.33, 71.72]	74.20 [71.20, 76.75]	64.70 [60.55, 69.10]	<0.001	nonnorm
Fertility (median [IQR])	3.70 [2.16, 5.59]	6.21 [5.35, 6.65]	3.09 [2.60, 3.83]	3.80 [2.87, 5.08]	1.81 [1.57, 2.00]	4.72 [3.71, 4.95]	<0.001	nonnorm
Population (median [IQR])	6.77 [2.02, 18.34]	7.51 [1.60, 13.37]	4.56 [0.64, 14.82]	12.21 [3.41, 55.63]	8.82 [4.30, 19.20]	0.24 [0.11, 2.73]	0.001	nonnorm
GDP (median [IQR])	8.22 [2.17, 51.77]	2.59 [1.02, 7.06]	7.96 [2.66, 37.18]	13.35 [4.89, 69.27]	71.95 [14.46, 193.92]	0.27 [0.18, 2.10]	<0.001	nonnorm

Summary

- In scientific journal articles, data is usually summarized in one table, referred to as “Table 1”
- The `tableone` package facilitates the creation of a “Table 1”
- The mean and standard deviation, or the median and IQR can be reported for continuous variables

Summarizing data

Reporting missing data

Heather Mattie, PhD

Missing data

Missing data is a common issue in biomedical datasets

- Can bias results and reduce statistical power
- Some models and functions will not run if there is missing data
- Clear communication of missingness improves reproducibility and transparency

gapminder example

```r
library(tidyverse)
library(dslabs)
data(gapminder)

df <- gapminder %>%
  filter(year == 1965)

head(df)
```

```
##               country year infant_mortality life_expectancy fertility
## 1             Albania 1965            94.10           66.59      5.59
## 2             Algeria 1965           149.10           50.09      7.66
## 3              Angola 1965               NA           38.74      7.43
## 4 Antigua and Barbuda 1965               NA           65.23      4.19
## 5           Argentina 1965            59.11           65.95      3.06
## 6             Armenia 1965               NA           69.26      3.98
##   population          gdp continent          region
## 1    1896125           NA    Europe Southern Europe
## 2   12626953  14486635612    Africa Northern Africa
## 3    5765025           NA    Africa   Middle Africa
## 4      59653           NA  Americas       Caribbean
## 5   22283389 130560392722  Americas   South America
## 6    2204650           NA      Asia    Western Asia
```

`is.na`

The **`is.na`** function returns TRUE if a value is NA (missing) and FALSE if it is not NA

- Using **`is.na`** with the **`sum`** function shows how many values are NAs in an object

```
sum(is.na(df))
```

```
## [1] 149
```

```
sum(is.na(df$country))
```

```
## [1] 0
```

```
sum(is.na(df$year))
```

```
## [1] 0
```

```
sum(is.na(df$infant_mortality))
```

```
## [1] 66
```

```
sum(is.na(df$life_expectancy))
```

```
## [1] 0
```

```
sum(is.na(df$fertility))
```

```
## [1] 0
```

```
sum(is.na(df$population))
```

```
## [1] 0
```

```
sum(is.na(df$gdp))
```

```
## [1] 83
```

```
sum(is.na(df$continent))
```

```
## [1] 0
```

```
sum(is.na(df$region))
```

```
## [1] 0
```

colSums

The **colSums** function with **is.na** returns the number of NAs in each column

```r
colSums(is.na(df))
```

```
##         country            year infant_mortality  life_expectancy
##               0               0               66                0
##       fertility      population              gdp        continent
##               0               0               83                0
##          region
##               0
```

rowSums

The **rowSums** function with **is.na** returns the number of NAs in each row

```r
rowSums(is.na(df))
```

```
##   [1] 1 0 2 2 0 2 2 0 0 2 1 1 0 0 2 0 1 0 2 0 2 0 0 2 1 0 0 2 0 0 2 0 1 0 0 0 2
##  [38] 1 0 0 0 2 1 2 2 0 2 0 0 0 0 2 2 2 2 0 0 0 1 1 1 1 2 0 0 2 2 0 1 2 0 1 0 1
##  [75] 0 0 0 0 0 1 1 1 0 1 0 1 2 0 1 0 1 2 2 1 1 0 0 1 2 0 2 2 0 0 0 1 1 1 0 1 0
## [112] 2 2 2 2 0 1 2 0 0 1 1 0 1 0 0 0 0 0 0 0 0 1 0 1 2 2 2 0 2 1 2 2 0 2 0 0
## [149] 0 2 2 1 1 0 0 0 2 1 0 1 0 2 1 0 2 0 1 0 0 0 2 1 2 1 0 0 0 2 1 0 2 1 1 0 0
```

`skimr` and `skim`

The `skimr` package provides compact and flexible ways to generate summary statistics

- The `skim` function outputs a comprehensive data summary

```r
library(skimr)
skim(df)
```

Data summary

Name	df
Number of rows	185
Number of columns	9
_______________________	
Column type frequency:	
factor	3
numeric	6
________________________	
Group variables	None

skimr and skim

Variable type: factor

skim_variable	n_missing	complete_rate	ordered	n_unique	top_counts
country	0	1	FALSE	185	Alb: 1, Alg: 1, Ang: 1, Ant: 1
continent	0	1	FALSE	5	Afr: 51, Asi: 47, Eur: 39, Ame: 36
region	0	1	FALSE	22	Wes: 18, Eas: 16, Wes: 16, Car: 13

Variable type: numeric

skim_variable	n_missing	complete_rate	mean	sd	p0	p25	p50	p75	p100	hist
year	0	1.00	1.965000e+03	0.000000e+00	1965.00	1.96500e+03	1.965000e+03	1.965000e+03	1.965000e+03	
infant_mortality	66	0.64	9.711000e+01	5.817000e+01	13.70	5.15500e+01	9.620000e+01	1.372000e+02	2.636000e+02	
life_expectancy	0	1.00	5.726000e+01	1.109000e+01	31.26	4.84400e+01	5.834000e+01	6.705000e+01	7.400000e+01	
fertility	0	1.00	5.380000e+00	1.830000e+00	1.81	3.60000e+00	6.070000e+00	6.800000e+00	8.200000e+00	
population	0	1.00	1.759520e+07	6.650895e+07	37963.00	7.53285e+05	3.559687e+06	9.504702e+06	7.065909e+08	
gdp	83	0.55	7.915212e+10	3.387851e+11	110094728.00	1.41424e+09	4.138937e+09	2.746349e+10	3.162743e+12	

`naniar`

The **`naniar`** package is a **`tidyverse`**-friendly package designed to facilitate the exploration, visualization, and manipulation of missing data.

- It provides a suite of tools for understanding the patterns and mechanisms of missingness within a dataset

gg_miss_var

The **gg_miss_var** function is used to visualize the missing data in a dataset by variable.

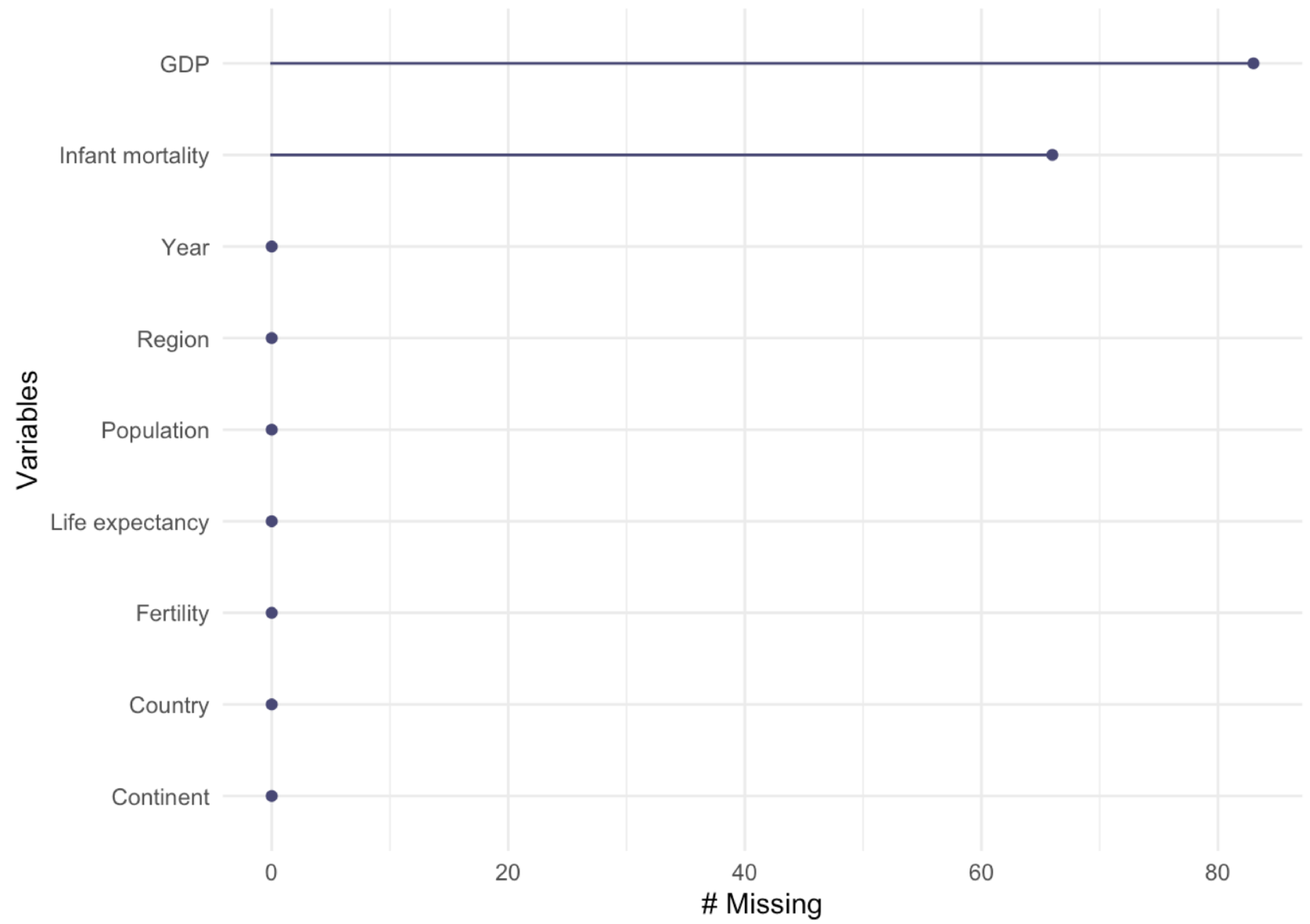
- Generates a **ggplot2** plot that displays the number of missing values for each variable (column) in a data frame

```
library(naniar)

df <- df %>%
  rename("Infant mortality" = infant_mortality,
         "Life expectancy" = life_expectancy,
         "Population" = population,
         "Fertility" = fertility,
         "Continent" = continent,
         "Country" = country,
         "Region" = region,
         "Year" = year,
         "GDP" = gdp
         )

gg_miss_var(df)
```

`gg_miss_var`

The `gg_miss_var` function is used to visualize the missing data in a dataset by variable.

- Generates a `ggplot2` plot that displays the number of missing values for each variable (column) in a data frame

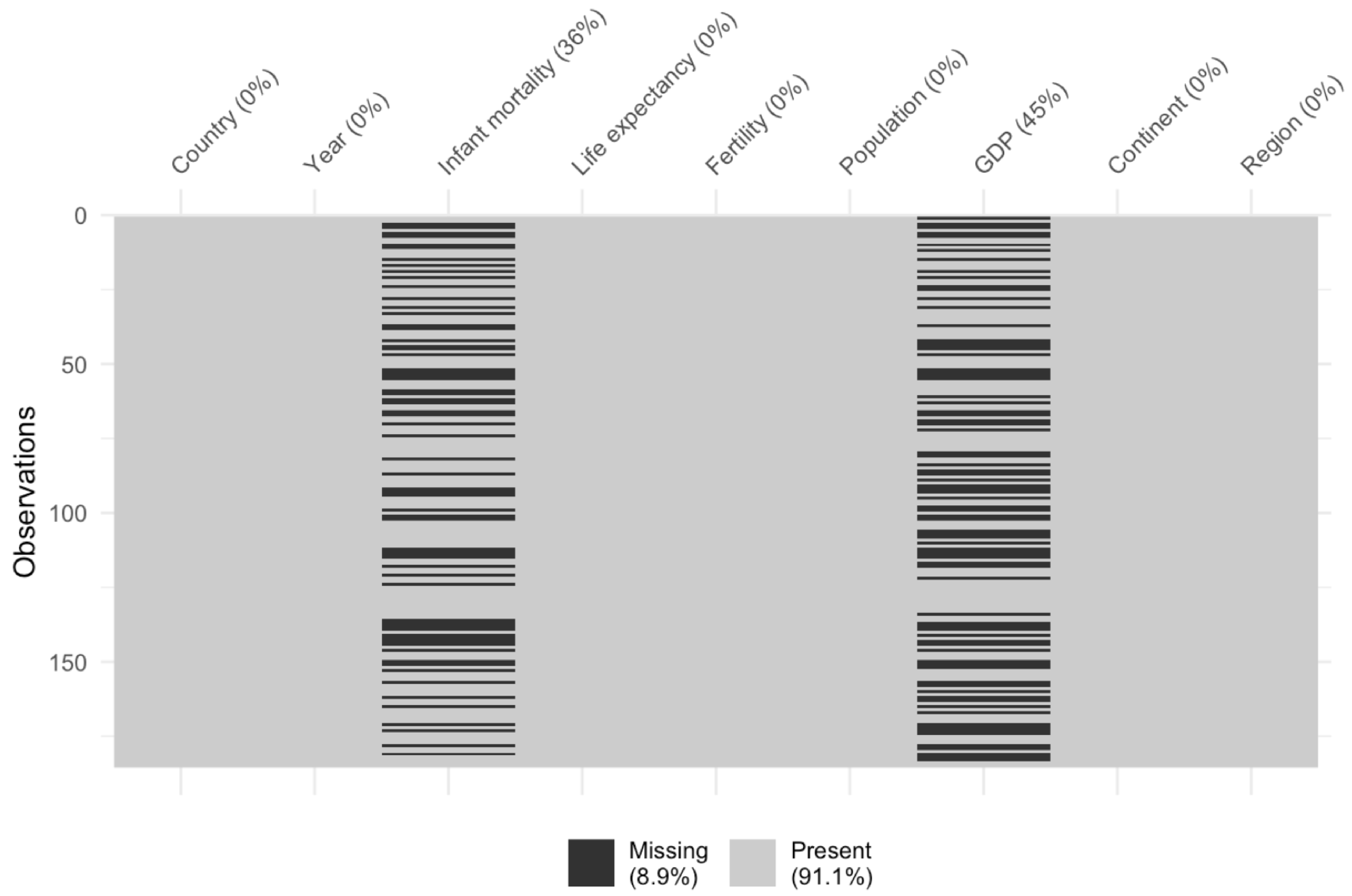


`vis_miss`

The `vis_miss` function allows for the visualization of missing data patterns within a data frame.

- Generates a `ggplot` object, offering a quick and visual overview of where missing values and present values are located within the dataset.

```
vis_miss(df)
```



Summary

- Missing data is a common issue and should be reported
- The `colSums` and `rowSums` can be used to show the number of missing values in columns and rows
- The `skim` function can be used to generate a comprehensive missing data summary
- Missing data can be visualized by using the `gg_miss_var` and `vis_miss` functions